

Luke Yamaguchi

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EDUCATION

University of California, Los Angeles (UCLA)	Sep 2024 - Jun 2027
M.S. Electrical & Computer Engineering (Exceptional Student Admission Program)	Expected Jun 2027
B.S. Computer Engineering	Expected Jun 2026
• GPA: 3.85/4.00 • Relevant Coursework: Computer Systems Architecture, Computer System Security, Digital Circuits, Operating Systems, Algorithms and Complexity, Data Communications and Telecommunication Networks, Communication Systems, Digital Signal Processing, Neural Networks and Deep Learning	

SKILLS

Languages: Verilog, SystemVerilog, C, C++, Python, Bash
Embedded Systems: RTL Design, FPGA (Artix 7), STM32, ARM Cortex-M7, UART, ADALM-Pluto SDR
Tools: Vivado, Git, Linux, Docker, GDB, MATLAB, Oscilloscopes, DMM, Analog Discovery 2
Language: English, Japanese

EXPERIENCE

Graduate ASIC/FPGA Security Engineering Intern	Los Angeles, CA
The Aerospace Corporation	Jun 2026 - Sep 2026
• Incoming summer 2026	
Undergraduate Researcher	Los Angeles, CA
Secure Systems and Architectures Lab - UCLA	Oct 2025 - Present
• Researching BLE hardware security and RF device authentication using PHY-layer hardware fingerprints • Built GNU Radio DSP pipeline to extract physical-layer features from BLE signals captured from ADALM-Pluto SDR • Trained ML-based RF fingerprinting models on Linux remote servers • Implementing adversarial RF spoofing attacks to evaluate authentication robustness under attacker-controlled interference	

PROJECTS

Hardware Security Lead — Mini-GPU FPGA Accelerator	Mar 2024 - Present
• Designed RTL communication/control path with RX/TX packet parsers, command decoding, and UART transport • Built SHA-256 kernel authentication to block execution of untrusted instruction kernels on FPGA • Integrated launch gating, program-memory locking, and ACK/NACK fault signaling in Verilog • Optimized SHA-256 RTL, reducing logic usage from ~10.5k to ~4.9k cells • Identified and patched security flaws including CRC-failed writes, invalid launch ACKs, and residual instruction memory after reset	
Hardware-Enforced Authentication System	Feb 2025 - Mar 2025
• Designed and implemented multi-user authentication system in Verilog on an Artix 7 FPGA • Implemented hardware-enforced role-based access control (RBAC) with admin, user, and guest privilege separation • Managed dynamic credential lifecycles, including creation, modification, & deletion • Engineered fail-secure features (auto re-locking & brute-force lockout) informed by authentication threat modeling • Validated logic integrity via .vcd waveform analysis and 670+ lines of simulation testbenches	
Project Lead & Software Lead — Mars Rover, 48-hour UCLA Hack Competition	Jul 2024
• Led 4-member team building Raspberry Pi Pico rover with sensors, ESP32 camera, and React web interface • Developed Python firmware for motion/data control, optimizing it to reduce MQTT communication latency by 86% • Earned 3rd place overall	
Lead Researcher — Multi-Agent Access Control	Oct 2025 - Dec 2025
• Proposed provenance-based access control framework to prevent Confused Deputy attacks in multi-agent LLM systems • Designed instruction-origin tainting policy for least-privilege enforcement • Reduced attack success rate from 57% to 29% across 15 test cases	